

Q33

Pollard's method :

find factor of $N = 91$

using $x_0 = 2$, $f(x) = x^2 - 1$

Remainder : $100 \equiv 9 \pmod{91}$

$$x_0 = 2$$

$$x_1 = f(2) = 2^2 - 1 = 3$$

$$x_2 = f(3) = 3^2 - 1 = 8$$

$$x_3 = f(8) = 8^2 - 1 = 63$$

$$x_4 = 63^2 - 1 = 3969 - 1$$

$$\equiv 3900 + 69 - 1$$

$$\equiv 39 \times 9 + 69 - 1$$

$$\equiv 400 + 20 - 1$$

$$\equiv 4 \times 9 + 20 - 1$$

$$\equiv 55$$

$$x_5 = 21$$

$$x_6 = 76$$

$\vdots$

$$\gcd(x_1 - x_0, 91) = 1$$

$$60 = 2^2 \cdot 3 \cdot 5$$

$$\gcd(x_2 - x_1, 91) = 1$$

$$\gcd(x_3 - x_1, 91) = 1$$

$$\gcd(x_4 - x_3, 91) = \gcd(-8, 91) = 1$$

$$\gcd(x_5 - x_3, 91) = \gcd(-42, 91) = \underline{\underline{7}} \text{ STOP}$$

Q 18

Alice's public key : $k_E = (N, e)$

" private " $= k_D = (N, d)$

Alice sends to her bank the :

$$\text{Signature} = m = (\text{message})^d \bmod N.$$

Bank reads

$$m^e \equiv \text{message} \bmod N.$$

Q 20 Alice $k_E = (n, e) = (713, 7)$

~~plaintext~~ $m = 37$

Cypher text =

$$37^d \text{ mod } 713$$

$$37^7 \text{ mod } 713$$

$$50653 = 37^3 \equiv 30 \text{ mod } 713$$

$$71 \times 713 = 50623$$

$$(37)^7 = (37^3)^2 \cdot 37$$

$$\equiv (30)^2 \cdot 37$$

$$\equiv 187 \cdot 37$$

$$\equiv 6919$$

$$\equiv 502 \text{ mod } 713$$

$$9 \times 713 = 6417$$

Q21 See lecture notes

Q22 A primality test typically establishes that a number is composite without exhibiting the factors.

Q23

An odd composite number m is
a pseudo-prime to base a

if $a^{m-1} \equiv 1 \pmod{m}$.

Is 91 a pseudo-prime to base 2?

$$2^{91-1} = 2^{90} \pmod{91}$$

$$\equiv 2^{72} \cdot 2^{18}$$

$$\equiv 2^{18}$$

$$\equiv 2^{10} \cdot 2^8$$

$$\equiv 1024 \cdot 256$$

$$\equiv \cancel{1024 + 24 + 2 \cdot 9 + 56}$$

$$\equiv \cancel{90}$$

$$\equiv (90 + 24)(18 + 56)$$

$$\equiv 23 \cdot 74 \not\equiv 1 \pmod{91}$$

$$\phi(91) = \phi(7 \cdot 13)$$

$$= \phi(7) \phi(13)$$

$$= 6 \cdot 12$$

$$\equiv 72$$

$$2^{\phi(91)} \equiv 1 \pmod{91}$$

91 is not a pseudo-prime
to base 2.

Is 91 a pseudo-prime to base 3?

$$3^{91-1} = 3^{90} = 9^{45}$$

$$= (9^2)^{22} \cdot 9$$

$$= (-10)^{22} \cdot 9$$

$$= (100)^{11} \cdot 9$$

$$= 9^{11} \cdot 9$$

$$= 9^{12} = (-10)^6$$

$$= (100)^3 = 9^3$$

$$\equiv -10 \cdot 9$$

$$\equiv -90$$

$$\equiv -(-1)$$

$$\equiv 1 \quad \text{mod } 91.$$

So 91 is a pseudo-prime to

base 3.

Tutorial : Moa 12, McMunn.